Preparing yourself and your laptop for Agentic AI classes

Last amended: 29th June. 2026
My folder: C:\Users\ashok\OneDrive\Documents\FORE_data_analytics_course\2026\agentic AI
Zoom the pdf file to 180% to see figures clearly.

Contents

1. Get API/Access-token for sites:

Our agents will need to work with or use several platforms. For this, we need API (HuggingFace calls it *Access Token*). Please get API/Access token for the following sites. You can record your API keys or token on a text file or in one of your emails for quick access in the class. While getting APIs/tokens **also get familiar** with the purpose of the site.

For all sites, creating an account with them is the first requirement. For most sites, you can use your FORE's gmail (fsm.ac.in) account to quickly log-in to create an account and then go on to get API key. For HuggingFace, you may use your gmail account but then you also have to write a password (that may be any password, not necessarily your gmail password) with which you intend to log-into this site in future and also, I think, the site requires you to specify & verify your mobile number.

Please note that this activity must be done at home. In class, when several persons log into a site (from a single IP), the sites resist and slow down and getting an API key takes very long time.

Where to save your APIs/Token

I generally keep these APIs and tokens on my laptop as also in my email. Email can easily be opened in the class. First, collect all of them, then write an email to yourself listing these APIs in it.

List of sites:

1. Cerebras API key
2. Pinecone API key
3. Huggingface—Generate access tokens for *Read Role*. Proceed as follows:

To get an **Access Token** (API Key) from Hugging Face, you can quickly **generate one from your account settings**. These tokens act as passwords to access private models, upload resources, or use the Inference API programmatically. Hugging Face +3

Here is a visual guide on how to safely store and use your Hugging Face API tokens in various development environments:

Step 1: Generate the Token

1. Go to Hugging Face and log into your account.
2. Click on your **Profile icon** in the top-right corner and select **Settings**.
3. In the left-hand menu, click on **Access Tokens**.
4. Click the **New token** (or **Create new token**) button.
5. Give your token a name, and select a **Role** (usually **Read** is sufficient unless you need to push code/data, in which case use **Write**).
6. Click **Generate a token**. Hugging Face +4

> **Important:** Copy and save your new token somewhere secure immediately. For security reasons, Hugging Face will not show it to you again once you close the page. YouTube · Amit Thinks +1

4. Serper API key
5. Serp API key
6. Tavily API key
7. Groq API key
8. BraveSearch API key
9. Free News API key
10. Polygon Financials API key
11. OpenRouter API key
12. Mistral Studio API key

Exercise 1 (self-study).

Here is an exercise for you. Fill up both columns for each one of the above platforms. Write the purpose in your own words.

Site name	Purpose of the site
Cerebras	

2. Install WSL Ubuntu on Windows 10 or Windows 11

Ubuntu on Windows has a special name: *WSL Ubuntu*.

Those of you who have a laptop with at the least i5 processor and 16gb RAM, may be able to perform some of the experiments on their laptops also. In case, your laptop configuration so allows, increase the RAM to 24gb or 32gb. RAM is relatively cheap and easy to plugin. Higher the RAM, better it will be.

All our software frameworks are installed on Ubuntu. We, therefore, need *WSL Ubuntu* OS for our software. Ubuntu can be installed *within* Windows. After Ubuntu is installed, we can install our needed frameworks. Ubuntu can be installed on your laptop along with your Windows 10 or Windows 11.

Install is easy and quick. Install WSL ubuntu on Windows 10/Windows 11 as per the following instructions. If you face any problems, please take the help of ChatGPT or Claude Desktop.

Follow these instructions:

> To install WSL and Ubuntu on Windows, open PowerShell as an Administrator, run `wsl --install`, and restart your computer. Once your PC reboots, the setup will automatically finish, launch the Ubuntu terminal, and prompt you to create a Linux user account. Medium · Siddheswar Samanta +1
>
> ## Step-by-Step Installation Guide
>
> **1. Install WSL**
>
> 1. Right-click the **Start menu** and select **Terminal (Admin)** or **PowerShell (Admin)**.
> 2. Type the following command and press Enter:
>
> powershell
>
> ```
> wsl --install
> ```
>
> Use code with caution.
>
> 3. This command enables the required Windows features, downloads the latest Linux kernel, and installs Ubuntu by default. Boot.dev +1
>
> **2. Restart Your Computer**
>
> - Wait for the installation to finish, then restart your PC to apply the necessary virtualization features. Medium · Siddheswar Samanta +1
>
> **3. Set Up Ubuntu**
>
> 1. After the reboot, search for **Ubuntu** in your Windows Start menu and open it.
> 2. The terminal will take a few moments to decompress and install.
> 3. Once prompted, enter a new **UNIX username** and **password**. LinuxConfig +1

This is how I do it on my laptop:

Right-click on Windows PowerShell and run it as *Administrator*. Execute the *wsl.exe* command as follows and press ENTER key.

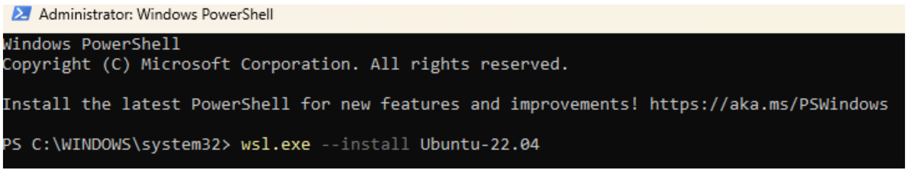
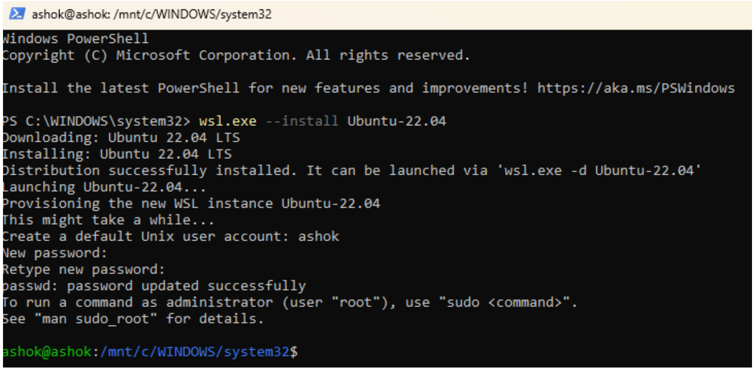


Figure 1: Installing WSL ubuntu 22.04 on Windows 10 or Windows 11. Open PowerShell as Administrator. Issue command: ***`wsl.exe --install Ubuntu-22.04`***

After download and installation of ubuntu, a user account name and its password is asked for. THIS PART IS IMPORTANT. Keep both simple. In our lab machines both user account name and password are 'ashok' and 'ashok'. You can follow the same convention. (see below screen).

Restart the Windows machine when done.



```
ashok@ashok: /mnt/c/WINDOWS/system32
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> wsl.exe --install Ubuntu-22.04
Downloading: Ubuntu 22.04 LTS
Installing: Ubuntu 22.04 LTS
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu-22.04'
Launching Ubuntu-22.04...
Provisioning the new WSL instance Ubuntu-22.04
This might take a while...
Create a default Unix user account: ashok
New password:
Retype new password:
passwd: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ashok@ashok:/mnt/c/WINDOWS/system32$
```

Figure 2: As ubuntu installs, you are asked user account name and password. Note that the typed password is NOT visible on screen for safety reasons.

In the lab machines, ubuntu **username** is '*ashok'* and **password** is also '*ashok'*. This is to keep it simple, uniform and easy to remember as also type. You can also follow the same convention. Should you FOREGT ubuntu password, recovery may be difficult and reinstallation will have to be done—a long procedure.

Quick Video

Here is a quick video from Microsoft for WSL ubuntu installation:

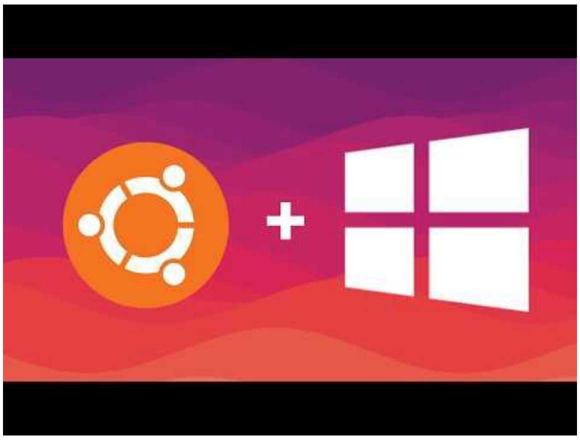


Figure 3: How to install ubuntu on Windows 10 or Windows 11

What next after WSL Ubuntu install

Once WSL ubuntu is installed, we can install our applications/frameworks. We have a script prepared for it. For running this script, I will send a separate email.

Exercise 2 (self-study):

Open WSL ubuntu and get familiar with the following Ubuntu commands. Take the help of ChatGPT or any other AI platform for the purpose. Remember that the sentence-case of the command is important. Generally small case is used.

File & Navigation Commands

- `pwd` — Prints the absolute path of your current directory.
- `ls` — Lists the files and folders inside your current location.
- `ls -la` — Shows detailed file listings, including hidden system files.
- `cd [path]` — Changes your working directory to the specified destination.
- `cd ..` — Moves your terminal session up one folder level.
- `mkdir [name]` — Creates a completely new directory folder.
- `touch [file]` — Creates an empty file or updates timestamps.
- `cp [src] [dest]` — Copies files from one location to another.
- `mv [src] [dest]` — Moves or renames files or system folders.
- `rm [file]` — Permanently deletes a specific file from the disk.
- `rm -r [folder]` — Recursively deletes a folder and all its contents. Dell +3

Figure 4: Ubuntu navigation commands. Learn them using ChatGPT or Claude Desktop

Next fill up the following table for each one of the above commands:

Command	Result-output or Your working example
pwd	/home/ashok
ls	
ls -la	

There will be a class-quiz on this.

3. Notepad++ install

Install notepad++ from here. Installation on Windows is straight forward, just click next..next..finish

4. Visual Studio Editor install

We will be working with python. We will use *Visual Studio Editor* for it. Install *Visual Studio Editor,* on Windows, as follows. Once installed, install python extension within it as explained in the following video BUT DO NOT install C++ extension—no need.

Figure 5: How to install Visual Studio Code Editor on Windows 10 or 11. Video link is here

5. Uninstall WSL Ubuntu

To uninstall *WSL Ubuntu*, start Windows PowerShell as Administrator and execute the command as follows and Restart the Windows:

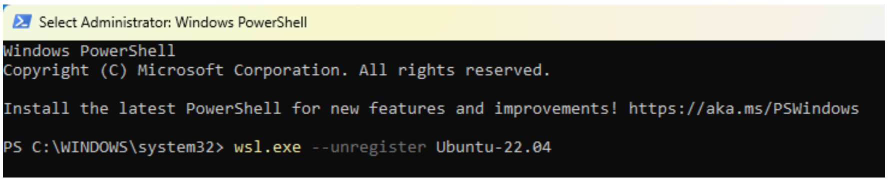


Figure 6: To uninstall WSL ubuntu, execute the command: `wsl.exe --unregister Ubuntu-22.04`

The result of this execution is as follows. Restart Windows thereafter.

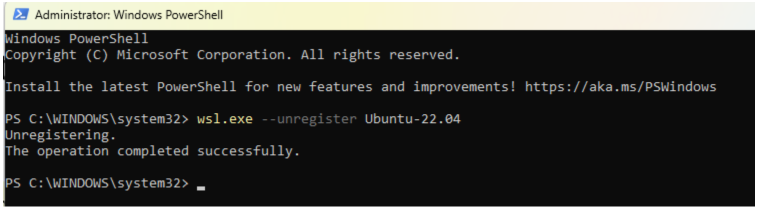


Figure 7: Result of execution. Restart Windows. Done...

6. Visual Studio Editor on WSL Ubuntu

Visual Studio Editor can be installed within WSL Ubuntu. One pre-requirement is that it should have been first installed on Windows. So before you come here, please follow the instructions as above to install VSCode in MS Widows.

Once done, open WSL Windows, and enter the code: `code    .` (code is followed by a 'dot') and press ENTER key (see below).

Figure 8: Write `code   .` *and press ENTER key to begin downloading and installing VScode in WSL Ubuntu.*

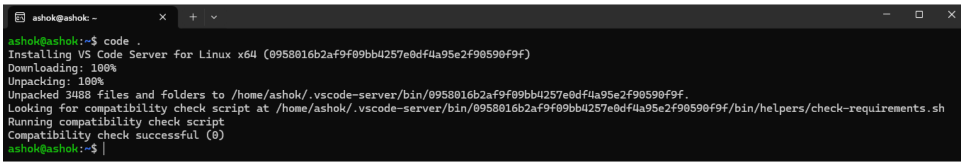


Figure 9: VScode downloaded in WSL Ubuntu

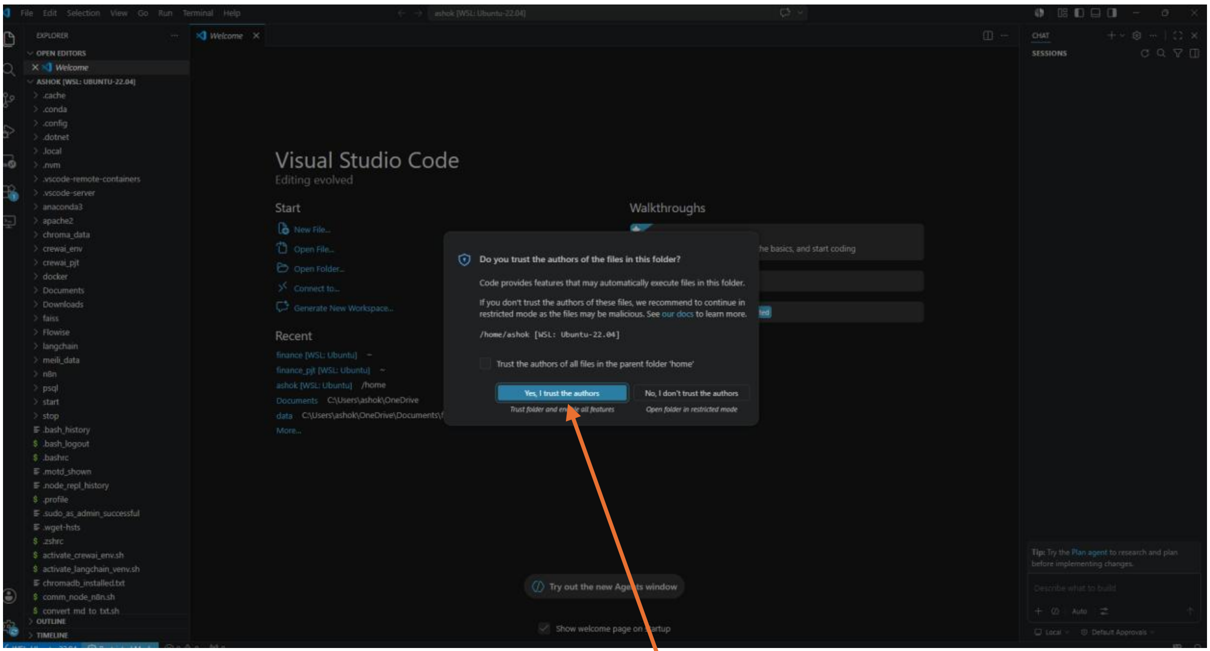


Figure 10: Click on Yes, I trust the authors button. Visual Studio Code opens.

Once done, you have two essential tasks: a) Install *WSL related* extensions, and b) install *python* related extensions. You have to use the *Extensions* icon:


Figure 11: One the left-most panel, this is the Extension icon for installing extensions of all sorts.

Click on this icon and in the search bar type: WSL and install the top-few WSL-related extensions. Once done, type: python

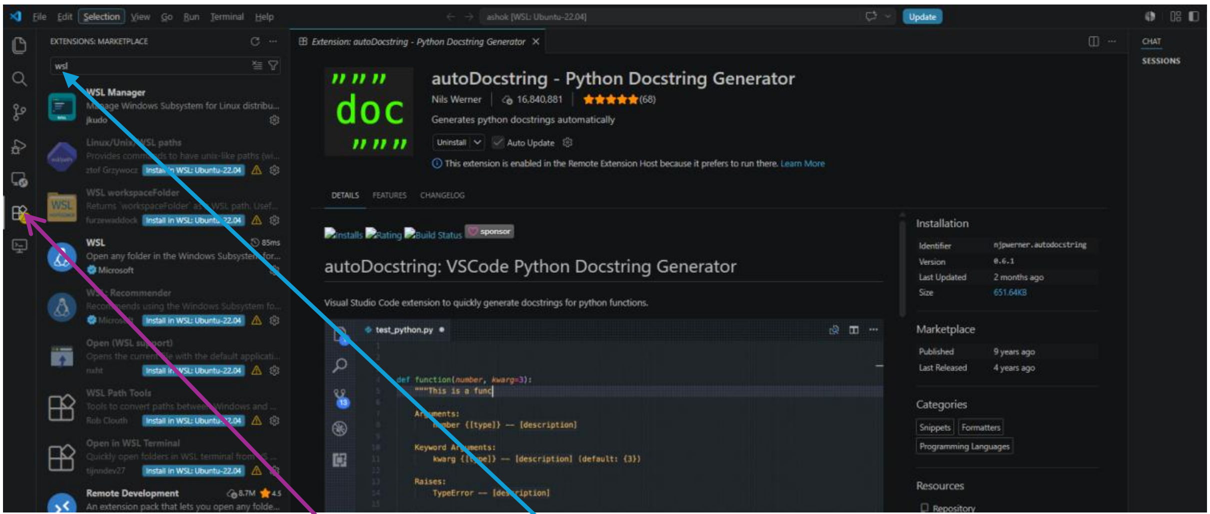


Figure 12: Click on Extension icon and type' wsl' in search bar. Then install top-few WSL related packages.

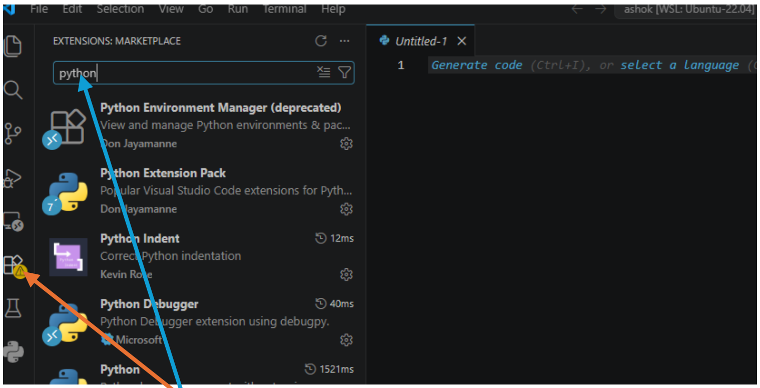


Figure 13: Look for python extensions to be installed and then install a top-few in WSL Ubuntu

Once WSL and python extensions are installed, we are finished.

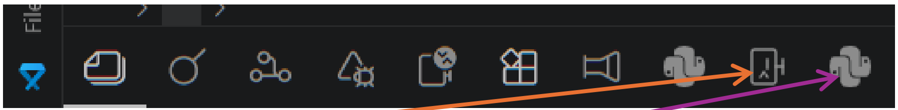


Figure 14: These two icons indicate that WSL Manager extension and python extensions have already been installed in my VSCode

Security permissions to WSL files

You may have to configure VScode's security settings. Open Ubuntu terminal. Open VScode in WSL ubuntu, type: **code .** (i.e. *code* followed by a *space* and then a *dot*) and press ENTER key. Follow these steps (see the two figures below):

1. With VScode open, click ctrl+, . This will open *Settings* menu.
2. Click *User* and then *Security*.
3. Search for *Allowed UNC*
4. Click *Add item* button, and add: wsl.localhost (see Ist figure below)
5. Next, click *Remote [WSL: Ubuntu-22.04]* and then *Security*.
6. Click *Add item* button, and add: wsl.localhost (see IInd figure below)

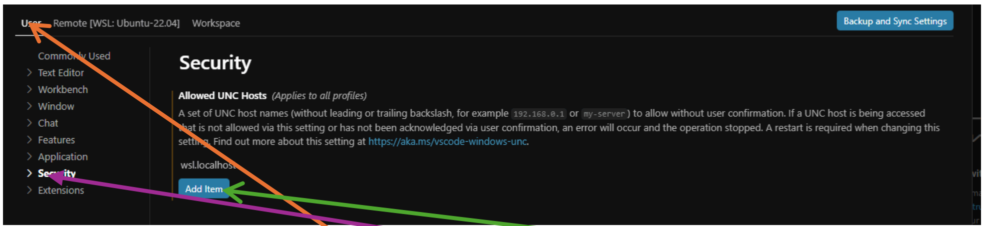


Figure 15: Click Ctrl+, to open Settings. Click User and then Security. Click Add item button to add: wsl.localhost

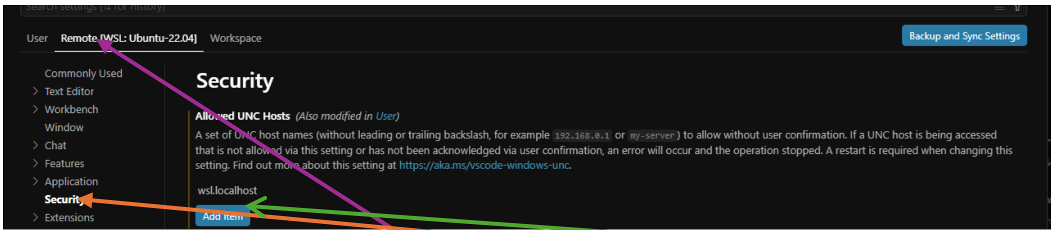


Figure 16: Click Ctrl+, to open Settings. Click *Remote [WSL: Ubuntu-22.04] and then Security. Add item: wsl.localhost*

Restart VSCode

7. Two factor authentication for your gmail account

Please enable two factor authentication in your *fsm* email account. We may need it while experimenting with n8n. To enable two-factor authentication, open google in *AI mode* (or open ChatGPT/Claude Desktop) and ask the question. *How to enable two factor authentication in gmail?* Follow the instructions. It is easy.

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